Exercises

Exercise 4.1 · Quadratics. Solve in $\mathbb{C}$: $z^2 + 9 = 0$; $z^2 - 4z + 13 = 0$; $2z^2 + 2z + 1 = 0$; $z^4 - 1 = 0$.

Exercise 4.2 · Reducing to a quadratic.

1. Solve in $\mathbb{C}$ the equation $z^4 + 4z^2 - 5 = 0$ by setting $Z = z^2$.
2. Solve in $\mathbb{C}$ the equation $z + \dfrac{1}{z} = 1$, for $z \neq 0$.

Exercise 4.3 · An obvious root. Let $P(z) = z^3 - 3z^2 + 4z - 2$.

1. Check that 1 is a root of P.
2. Determine the polynomial Q such that $P(z) = (z - 1)Q(z)$.
3. Solve $P(z) = 0$ in $\mathbb{C}$.

Exercise 4.4 · A non-real root. Let $P(z) = z^3 + z^2 + 4z + 4$.

1. Check that $2i$ is a root of P. Deduce another root without computation.
2. Expand $(z - 2i)(z + 2i)$, then factorise $P(z)$ and solve $P(z) = 0$.

Exercise 4.5 · Factorising $z^3 - 27$.

1. Factorise $z^3 - 27$ by $z - 3$ using the course.
2. Solve in $\mathbb{C}$ the equation $z^3 = 27$ and plot the solutions in the complex plane. What do you notice?

Exercise 4.6 · A parameter. Let m be a real number. Consider the equation (E_m): $z^2 - 2z + m = 0$.

1. Depending on the value of m, give the number and the nature of the solutions of (E_m).
2. Assume $m > 1$. Let A and B be the images of the solutions. Determine m so that the triangle OAB is equilateral.

Exercise 4.7 · Recovering a polynomial. Determine the polynomial P of degree 3, with real coefficients and leading coefficient 1, such that $1 + i$ is a root of P and $P(0) = 4$.